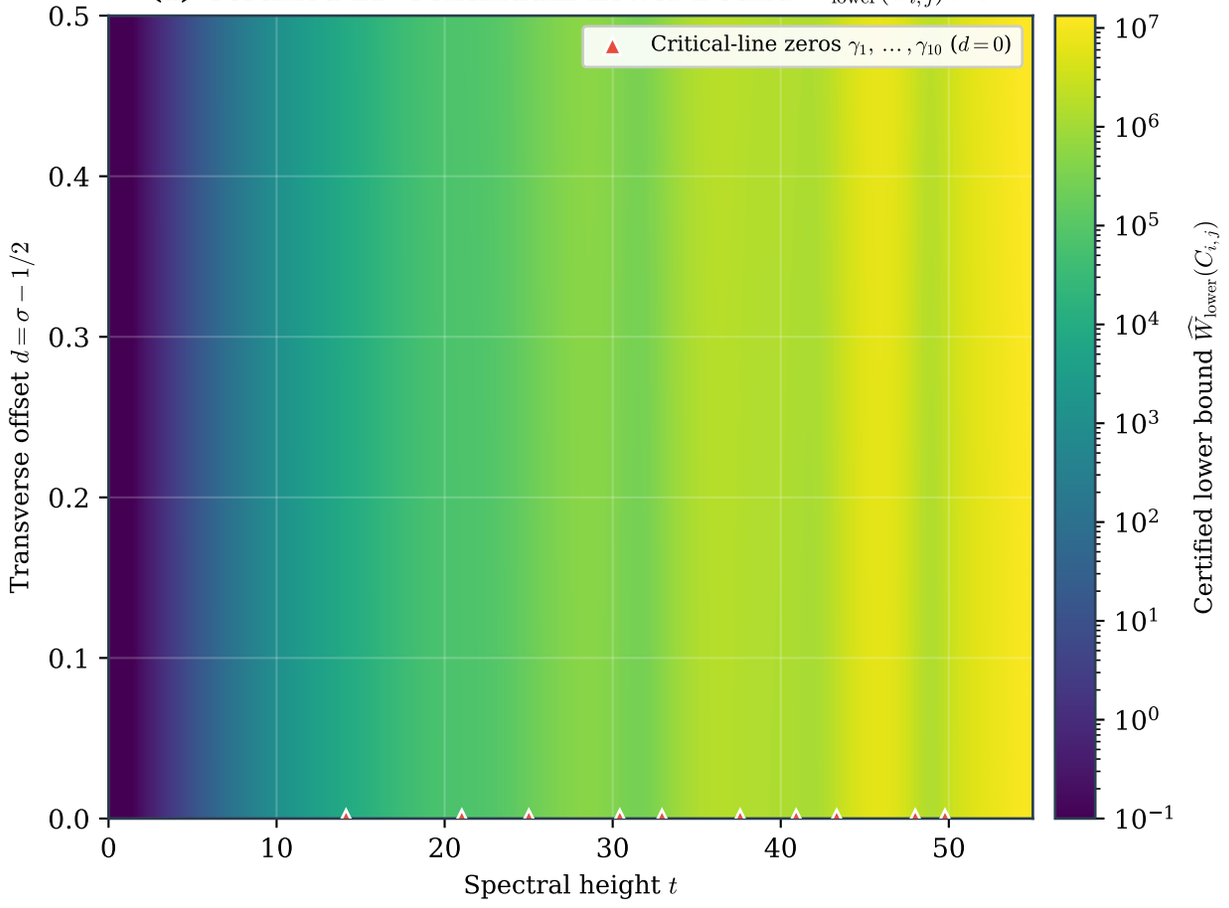
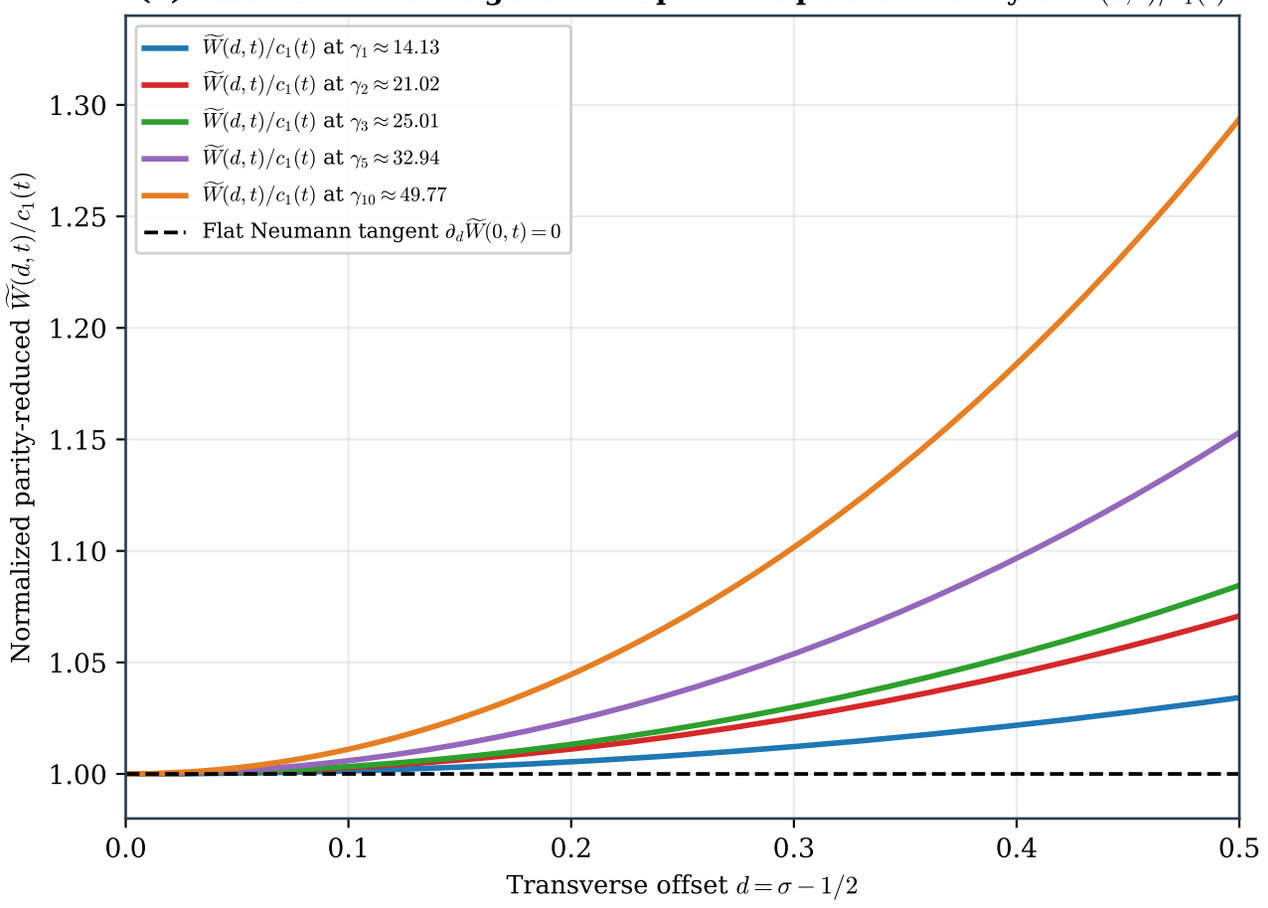
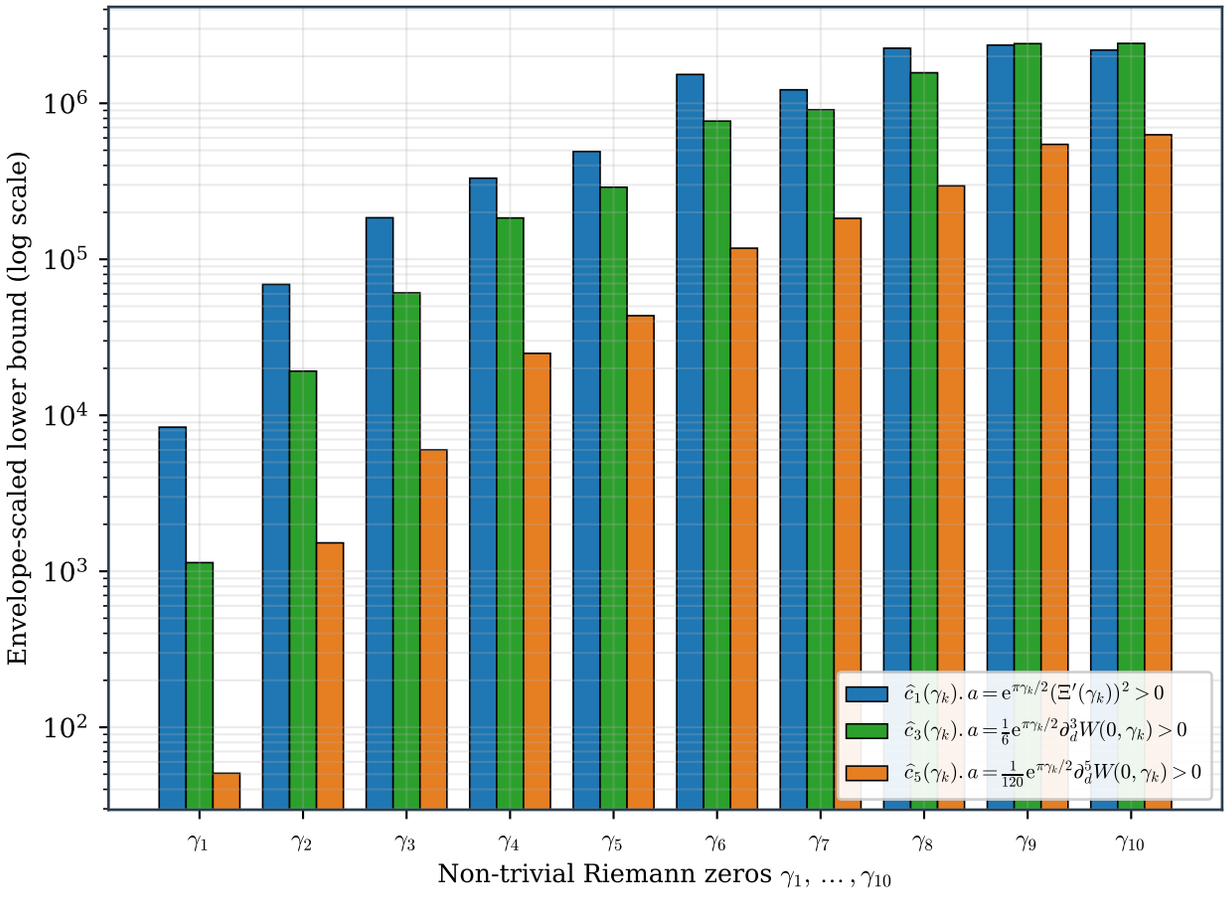
(a) Certified 2D Continuum Lower Bound $\widehat{W}_{\text{lower}}(C_{i,j}) > 0$



(b) Flat Neumann Tangent and Upward Dipole Convexity of $\widetilde{W}(d,t)/c_1(t)$



(c) Interval-Certified Dipole Invariants $\widehat{c}_1, \widehat{c}_3, \widehat{c}_5 > 0$ at $\gamma_1, \ldots, \gamma_{10}$



(d) Critical-Line Boundary Invariant $\widetilde{W}(0,t) = c_1(t) > 0$ $(d=0)$

